

Certamen №2

(**Date:** June 03, 2025)

1. (**7 points**) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r, t) = \frac{k(t) r^2}{2}, \quad k(t) = \frac{\kappa}{t^2}, \quad \kappa > 0. \quad (1)$$

- (a) (**2 points**) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta = \text{const}$$

which do not change the equations of motion.

- (b) (**3 points**) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion. Note that the lagrangian of the 3D harmonic oscillator in Cartesian coordinates can be represented as a sum of 3 independent pieces,

$$L = \sum_{a=1}^3 \left(\frac{m \dot{x}_a^2}{2} - \frac{k(t) x_a^2}{2} \right),$$

so in Cartesian coordinates the equations of motion decouple, and may be analyzed independently for each component.

- (c) (**2 points**) We have seen in lectures that in static (time-independent) spherical harmonic oscillator the particle's trajectory is a plain ellipse. Analyze if this property remains valid in the time-dependent potential given in Eq. (1).
2. (**8 points**) In the lectures 12-13 we discussed that the covariant equations of motion for the massive particle in the electromagnetic field A^μ may be obtained from the action

$$S = \int d\tau \Lambda, \quad \Lambda = -mc^2 \sqrt{\dot{x}_\mu \dot{x}^\mu} + e \dot{x}_\mu A^\mu(x) \quad (2)$$

where τ is the proper time, $x^\mu(\tau)$ is the trajectory of the particle, and $\dot{x}^\mu = dx^\mu/d\tau \equiv u^\mu$ is the 4-velocity of the particle. Assume that we replaced the covariant lagrangian (2) with

$$\tilde{\Lambda} = \frac{mc^2 \dot{x}_\mu \dot{x}^\mu}{2} + e \dot{x}_\mu A^\mu(x) \quad (3)$$

- (a) (**2 points**) Demonstrate that the action S is covariant under transformations of the Lorentz group when we use the lagrangian (3).
- (b) (**2 points**) Write out the generalized momentum $\mathcal{P}_\mu = \partial \tilde{\Lambda} / \partial \dot{x}^\mu$ and the equations of motion which correspond to the lagrangian (3).
- (c) (**2 points**) Demonstrate that your equations of motion are equivalent to the equations of motion found in lectures 12,13 using the lagrangian (2). Demonstrate that the same equations can be obtained with any lagrangian $\bar{\Lambda}$ given by

$$\bar{\Lambda} = mc^2 f(\dot{x}_\mu \dot{x}^\mu) + e \dot{x}_\mu A^\mu(x)$$

where $f(y)$ is an arbitrary function which satisfies $\partial f(y)/\partial y = 1/2$ at $y = 1$ [clearly, the lagrangians (2,3) correspond to $f(y) = \sqrt{y}$ and $f(y) = y/2$, respectively].

- (d) (**2 points**) Find the nonrelativistic lagrangian L which corresponds to the lagrangian (3). Pay attention that the nonrelativistic lagrangian L contributes to the action as $S = \int dt L$.